



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science & Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT2142
Final Examination
Date: 3/20/2024

Course Title: Ordinary and Partial Differential Equations
Semester: Fall 2023
Total Marks: 40

Total Time: 2 Hours

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) State the Lagrange's method [2]
(b) Solve the following problems by using Lagrange's method:
(i) $p \tan x + q \tan y = \tan z$. [4]
(ii) $y^2 zp - x^2 zq = -x^2 y$. [4]
2. Solve the following problems:
(i) $(D^2 - 4DD' + D'^2)z = x^3 y^3$. [5]
(ii) $(D^2 + 5DD' + 2D'^2)z = e^{2x+5y}$. [5]
3. Find the complete integral of the following differential equations:
(i) $\frac{x(1+p^2)}{1+y^2} + py(z - px)^2 - q = 0$. [5]
(ii) $(x^2 - y^2)pq - xy(p^2 - q^2) = 0$. [5]
4. (a) Write the working rule of the Charpit's method to find the complete integral. [3]
(b) Find the singular and general solution of $2xz + px^2 - 2qxy + pq = 0$. [7]
5. (a) Write the working rule of parabolic equation. [2]
(b) Reduce the equation $x^2 r - y^2 t = 0$ into canonical form. [8]
6. Define boundary value problem. Solve the following BVP: [10]
 $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$; $u = u(x, t)$; $0 < x < 1$; $t > 0$ and the conditions are $u(0, t) = u(1, t) = 0$; $u(0, t) = 3 \sin 3\pi$.
7. Solve the following problems:
(i) $(D^2 - D'^2 - D' - D)z = \sin(x + 3y)$. [5]
(ii) $(D^2 - D'^2)z = x^2 - y^2$. [5]



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering (CSE)
B.Sc. in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT213
Final Examination
Date: 3/18/2024

Total Time: 2:0 hours

Course Title: Statistics and Probability
Semester: Fall 2023
Total Marks: 40

Answer any four(04) of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a) Define probability, trail, outcomes, and events. [03]
- (b) A sample of four electronic components is taken from the output of a production line. The probabilities of the various outcomes are calculated to be: $P[0 \text{ defectives}] = 0.6561$, $P[1 \text{ defectives}] = 0.2916$, $P[2 \text{ defectives}] = 0.0486$, $P[3 \text{ defectives}] = 0.0036$, $P[4 \text{ defectives}] = 0.0001$. What is the probability of at least one defectives? [03]
- (c) Let X be a continuous random variable with the probability density function : [04]

$$f(x) = \frac{6}{8}x(2-x), 0 \leq x \leq 2$$
 Find (i) $P[X < 1]$
 (ii) $P[X > 1]$
 (iii) $P[0.5 < X < 1]$
2. (a) Define density and marginal density function. [02]
- (b) Let X and Y have the following distribution : [03]

$$f(x, y) = \begin{cases} x^2 + \frac{xy}{3}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$
 Check that $f(x, y)$ is a density function.
- (c) Find the marginal densities of X and Y from the following joint density functions and verifies that marginal distributions are also probability distribution, [05]

$$f(x, y) = \begin{cases} \frac{1}{8}(6-x-y), & 0 \leq x \leq 2, 2 \leq y \leq 4 \\ 0, & \text{otherwise} \end{cases}$$
 Also compute $P(X+Y < 3)$ and $P(X < 1.5, Y < 2.5)$
3. (a) Define continuous random variable. [01]
- (b) Calculate the mean, variance and the standard deviations for random variable sales from the following information provided by a sales manager of a certain business unit for a new product : [03]

Monthly sales(in unit)	50	100	150	200	250	300
Probability	0.10	0.30	0.30	0.15	0.10	0.05
- (c) The probability density function of a random variable x has the following [06]

$$f(x) = \begin{cases} \frac{x}{50}, & 0 < x < 10 \\ 0, & \text{otherwise} \end{cases}$$
 Find $E(x)$ and $V(x)$.
4. (a) Write a short note about binomial distribution. [02]



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(b) If $X \sim \text{binomial}(n, p)$ then show that,

(i) $M_x(t) = (pe^t + q)^n$

(ii) Mean = np , and Variance = npq

[04]

(c) About 30% items of the production bulbs are defective in a factory. 10 bulbs are packed in one packet. Among the 1000 packets find the probability that – (i) no bulbs is defective (ii) at least one bulbs defective (iii) at best one bulb is defective and find such kind of expected number of packets?

[04]

5. (a) Define poisson distribution.

[01]

(b) Show that for poisson distribution, Mean = Variance = equal.

[05]

(c) The probability that a person suffers from AIDS is 0.003. Let x be the number of persons who suffer in a sample of 1000 patients. Find (i) $P(x < 2)$, (ii) $P(x \geq 3)$, (iii) $P(2 \leq x \leq 4)$.

[04]

6. (a) Define normal distribution.

(b) Discuss about characteristics and importance of normal distribution.

[01]

(c) A recent study of the hourly wages of maintains crews for major airlines showed that the mean hour salary was TK. 165, with standard deviation of TK. 35. If we select a crew member at random, what is the probability the crew member earns:
Between TK. 165 and TK. 200 per hour?

[04]

More than 200 per hour?

[05]

Less than TK. 150 per hour?

2, 1 3, 4. 2, 3, 4. 4, 3, 2, 1

$$\frac{e^{-\lambda} \lambda^x}{x!}$$

1, 2, 3

0, 1



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE215: Data Structures

Batch: 23rd

Date: 16/03/2024

Final Examination

Semester: Fall 2023

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings:

5x8= 40

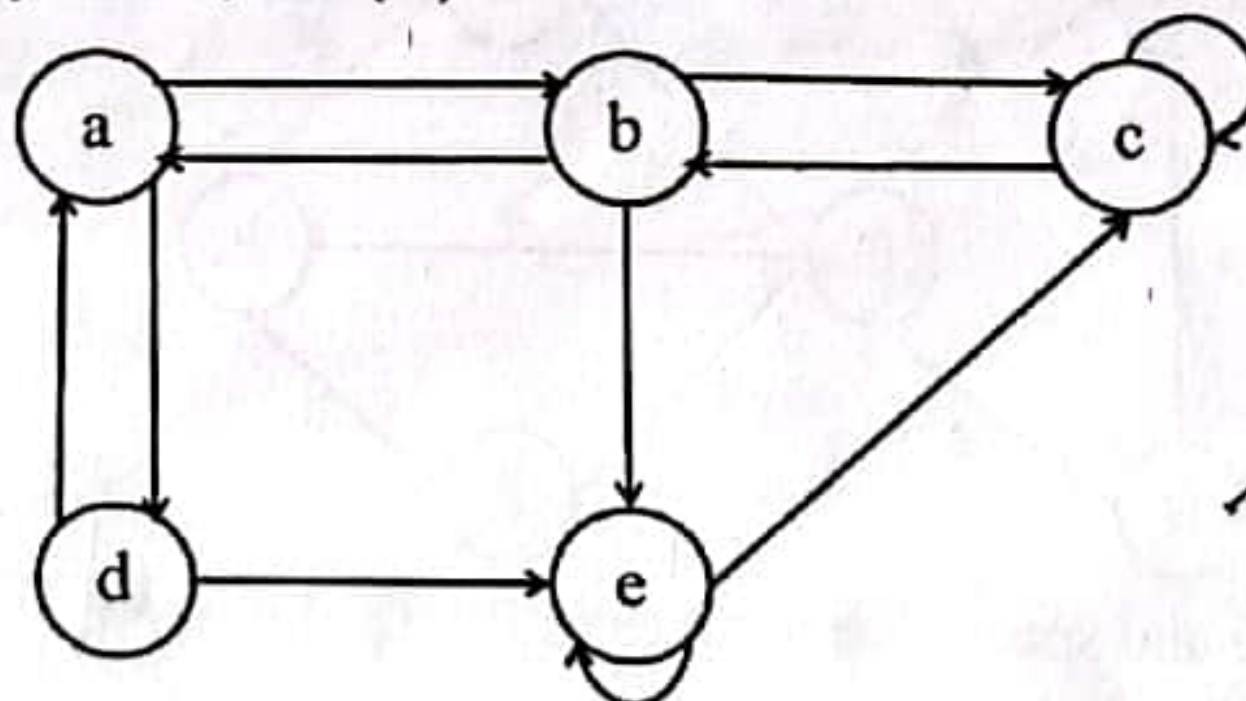
- 1 (a) Describe the Huffman algorithm used for compression. 3
(b) A string with different characters, say "bookkeeper". Find the Huffman code and build the Huffman tree for this string. 5

- 2 (a) State the differences between binary tree and complete binary tree. 3
(b) Determine the consistency of the following linear equations: 5
 $x + y + z = 3$
 $2x + 2y + 2z = 6$

- 3 (a) Perform the meet and join operations for the following two matrices: 2

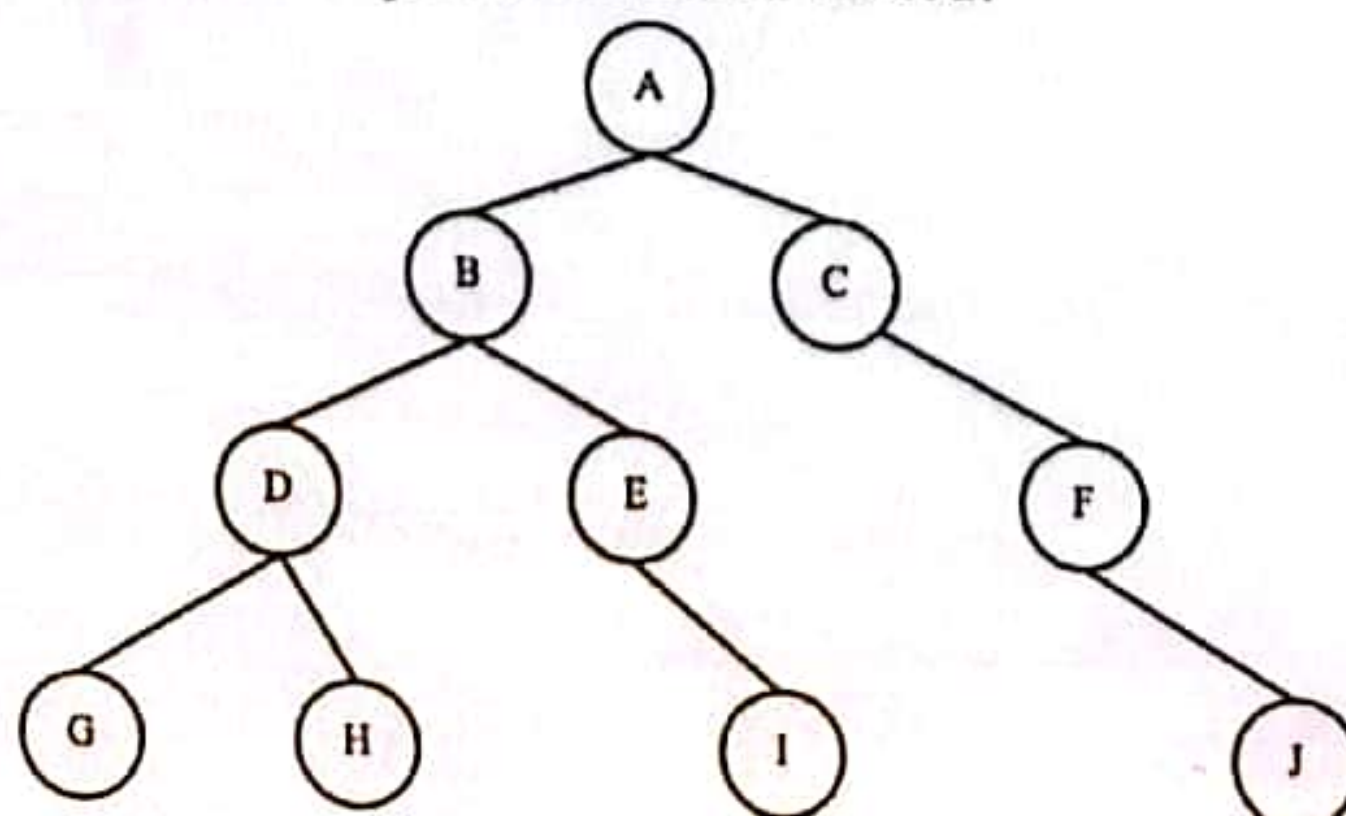
Matrix A:	Matrix B:
1 0 1	0 1 0
0 1 0	1 0 1
1 0 1	0 1 0

- 4 (a) Given the following graph, represent them using (i) adjacency list, (ii) adjacency matrix, and (iii) incidence matrix: 6



- 4 (a) Draw a binary tree to represent the algebraic expression $H = ((s / t) + (u * v)) - (w + x) * y$. 2

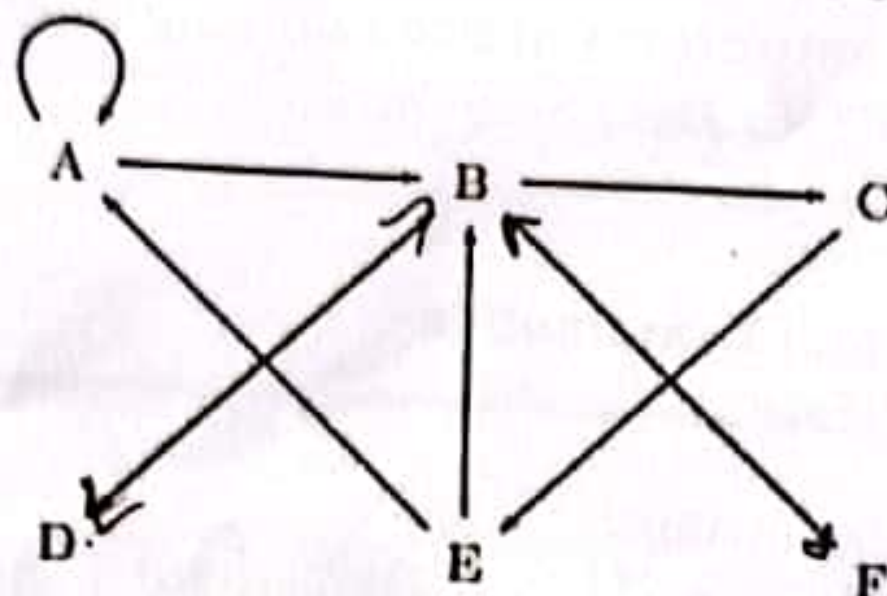
- (b) Perform pre-order, in-order and post-order traversal of the following binary tree and list the sequence of nodes visited: 3



P = R₀, L, R
In = L, R₀, R
Pos = L, R, R₀

- (c) Count the in-degree and out-degree of the following graph:

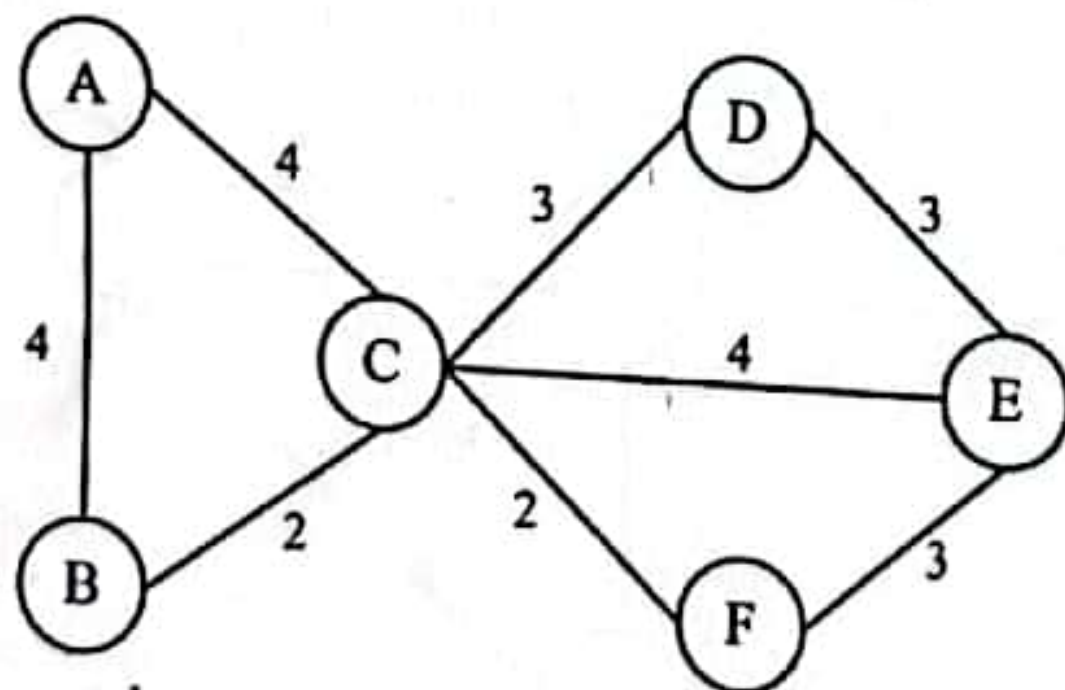
3



- 5 (a) List out the properties of Binary Search Tree (BST). 2
 (b) Analyze the complexity of Searching a Node from BST. 2
 (c) Describe the algorithmic approach to delete a node from BST. 4

- 6 (a) Define heap. 2
 (b) Identify the variants of heap structures. 2
 (c) Given the sequence of numbers 5, 3, 17, 10, 84, 19, 6, and 22, perform insertion using a max heap and illustrate the final heap structure. 4

- 7 (a) Define minimum spanning tree. 2
 (b) Build a minimum spanning tree using Kruskal's algorithm for the following graph: 3



- (c) Describe the time and space complexities of Prim's algorithm.

3

max = $\sqrt{0.75}$

Total Marks: 40

The figure in the right margin indicates full marks. Answer any Four (4) of the following questions:

1. Han Solo Consulting opened for business on December 1, 2023. Its trial balance before adjustment on December 31 is as follows: 10

HAN SOLO CONSULTING
Trial Balance
December 31, 2023

Titles of Accounts	Tk.	Tk.
Cash	29,600	
Supplies	2,300	
Prepaid Insurance	5,000	
Land	22,000	
Cottages	1,20,000	
Furniture	26,000	
Accounts Payable		5,500
Unearned Rent		8,400
Mortgage payable		60,000
P. Orbis, Capital		1,20,000
P. Orbis, Drawing	5,000	
Rent Revenue		80,000
Repair Expense	3,600	
Salaries Expense	51,000	
Utilities Expense	9,400	
	<u>2,73,900</u>	<u>2,73,900</u>

Other data:

- Insurance expires at the rate of Tk.1, 000 per month.
- A count of supplies shows Tk.1, 200 of unused supplies on December 31.
- Unearned rent of Tk.4, 100 was earned.
- Salaries Tk.400 are accrued and unpaid at December 31.

271900
271900



v) The Mortgage interest rate is 12%. (The Mortgage was taken out on December 1).

Requirement:

- Make adjusting entries at the end of the month of December 31, 2022.
- Prepare adjusted trial Balance on December 31, 2022.

2. Judi Dench is a licensed CPA. During the first month of operations of her business, the following events and transactions occurred:

10

April 1 Judi invested Tk.25,000 cash

April 2 Hired a secretary receptionist at a salary of Tk.1200 per month.

April 3 Purchased Tk.1,700 of supplies on account from Read supply Company.

April 7 Paid office rent of Tk.900 cash for the month.

April 11 Completed a tax assignment and billed client Tk.2,100 for services provided.

April 12 Received Tk.3,500 advance on a management consulting engagement.

April 17 Received cash of Tk.1,200 for services completed for H. Arnold co.

April 31 Paid Secretary receptionist Tk.1200 salary for the month.

April 31 Paid 60% of balance due Read Supply Company.

Requirement: Transaction post into the Ledger.

3. Tarek & Co., manufactures and sells a specialized cordless telephone for high electromagnetic radiation environments. The company's contribution format income statement for the most recent year is given below: (5*2=10)

Particular	Total	Per unit	% of sales
Sales (21,000 units)	12,60,000	Tk.60	100%
Variable expenses	<u>9,45,000</u>	<u>Tk.45</u>	<u>? %</u>
Contribution margin	3,15,000	<u>Tk.15</u>	<u>? %</u>
Fixed expenses	<u>2,40,000</u>		
Net operating income	<u><u>75,000</u></u>		

Management is anxious to increase the company's profit and has asked for an analysis of a number of items.

You are required to:

- Compute the company's CM ratio and variable expenses ratio.
- Compute the company's break- even point in both sales revenue and units.

iii) Assur
remain
iv) A
ea

- iii) Assume that sales increased by Tk.4,50,000 next year. If cost behavior patterns remain unchanged, by how much will the company's net operating income increase?
- iv) As per the original data, Assume that next year management wants the company to earn a profit of at least Tk. 1,20,000. How many units will have to be sold to meet this target profit?
- v) Refers to the original data, compute the company's margin of safety in both amount and percentage.

4. At the end of first month of operations, Bond Answering Service has the following unadjusted trial balance: 10

Bond Answering Service

Trial Balance

December 31, 2023

Titles of Accounts	Tk.	Tk.
Cash	5,400	
Accounts Receivable	2,800	
Prepaid Insurance	2,400	
Supplies	1,300	
Equipment	60,000	
Notes payable		40,000
Accounts Payable		2,400
Hambone, Capital		30,000
Hambone, Drawing	1,000	
Service Revenue		4,900
Salaries Expenses	3,200	
Utilities Expenses	800	
Advertising Expense	400	
	<u>77,300</u>	<u>77,300</u>

Other data:

- Insurance expires at the rate of Tk.400 per month.
- There are Tk. 500 of supplies on hand December 31, 2023.
- Depreciation on the equipment is Tk.900.



iv) Interest of Tk.500 on the notes payable has accrued during December.

Requirement:

- i) Prepare adjusting entries at the end of 31st December 2023.
- ii) Complete a Worksheet with 10 columns.

5. Write short notes (Any four):

2.5x4=10

- a) Assumption of CVP;
- b) Margin of Safety;
- c) Error in trial balance;
- d) Accounting cycle; and
- e) Application of CVP.



FACULTY OF SCIENCE AND TECHNOLOGY

BAN 211: History of emergence of Bangladesh

Semester: Fall 2023

12/03/2024

Final Examination 2024

Time: 2 Hours

Total Marks: 10x4=40

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any Four (4) of the following questions.

1 Table-1

Head of Comparison	West Pakistan	East Pakistan
Army Officers	95%	5%
Soldiers	96%	4%
High Officials in Navy	81%	19%
Other Post in Navy	91%	9%

Table-2

Years	Allocation for West Pakistan	Allocation for East Pakistan
1955-1960	Taka 500 Crore	Taka 113 Crore
1960-1965	Taka 22,230 Crore	Taka 6480 Crore

- a) Explain the discrimination between East and West Pakistan reflected in the table no. 1 05
- b) Evaluate the economic condition resulted from the discrimination between East and West Pakistan shown in the table no. 2. 05

- 2 a) Discuss about Tamaddun Majlish. 02
- b) Who are language martyrs? 03
- c) Discuss the role of the language movement 1952 in the evolution of Bengali Nationalism. 05

- 3 a) Write about Abdul Hamid Khan Bhashani, Sher e Bangla A K Fazlul Hoque, Huseyn Shaheed Suhrawardy. 05
- b) How many sectors were in Bangladesh during the Liberation War? Which sector do you belong to? Write about the sector in the Bangladesh liberation war? 05

- 4 a) "The struggle this time, is a struggle for our liberty. The struggle this time, is a struggle for our independence" in the speech, Bangabandhu informally declared independence of Bangladesh. 05
- b) What was the highlight of the 7th March of speech of Bangabandhu Sheikh Mujib? 05

CLO3

CLO3

CLO5

CLO5



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5	a) What was operation searchlight? Discuss about the <u>killing</u> of the intellectuals in the war of liberation of <u>Bangladesh</u> . b) Write down the role of India in the liberation war of Bangladesh.	05 05	CLO6
6	a) How was the first government of Bangladesh formed? 5 b) Discuss the role of national four leaders of Mujibnagar Government in the war of liberation.	05 05	CLO7

East



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics & Pulse Technique

Batch: 23rd
Date: 06/03/2024

Final Examination

Semester: Fall 2023
Time: 2 Hours
Total Marks : 40

Answer any five questions.

5 × 8 = 40

1. (a) Design a full adder and explain its operation. 4
(b) Explain the operation of a 4-bit ripple counter with timing diagram. 4

2. (a) Distinguish between asynchronous and synchronous counters. 4
(b) Sketch a circuit diagram for parallel in-parallel out register. 4

3. (a) List the commonly used IC packages. 4
(b) Draw a MOD-16 up-down counter. 4

4. (a) Illustrate a one-line-to-eight-line de-multiplexer. 4
(b) Sketch a BCD-to-Decimal decoder. 4

5. (a) Draw and explain analog-to-digital and digital-to-analog conversion. 4
(b) Write the steps of a small-project management. 4

6. (a) List the types of ROMs and explain any one of them. 4
(b) Explain the Read and Write operation from CPU. 4

7. Write short notes on:

4 × 2 = 8

- I. Static RAM
- II. Access time
- III. Auxiliary memory
- IV. Capacity